

DeCovarT reproducibility

Simulations, gastruloid GRNs, and sensitivity analyses

Bastien Chassagnol

2026-09-17

Table of contents

1	Overview	3
1.1	How to render	3
1.2	Scripts (source of truth for computation)	5
2	Suppinger gastruloids	6
3	Data access	7
4	Exploration template	8
5	Outputs	9
6	Gene regulatory networks	10
7	Graphical lasso per cell type	11
8	Feature selection	12
9	Pseudo-bulk markers	13
10	Genetic / Pareto panels	14
11	Sensitivity analyses	15
12	Contaminating Ω_j	16

1 Overview

This Quarto **book** is the narrative companion to the analysis scripts under `scripts/`. R chunks stay `eval: false` until artefacts exist. Mermaid and static Markdown still execute; `freeze: auto` caches those results in `docs/_freeze`.

The layout follows analysis reproducibility repos such as [cellrank2_reproducibility](#) (Python + Jupyter Book there). Here the stack is **R**, **renv**, and **Quarto**, published with the GitHub Pages `gh-pages` workflow ([Quarto: GitHub Pages](#), [Knitr with renv](#)).

```
%%{init: {"theme": "sandstone"}}%%  
flowchart TD  
  A["GEO GSE229513 / GSE229386"] --> B["02 preprocess SE / SCE"]  
  B --> C["03 glasso GRN per type"]  
  B --> D["04a pseudo-bulk DESeq2"]  
  D --> E["04b NSGA-II / GA panel"]  
  C --> F["DeCovarT deconvolution"]  
  E --> F  
  F --> G["05 contaminate Omega_j"]
```

1.1 How to render

From the repository root, after `renv::restore()`:

```
quarto render docs  
# first time only, once gh-pages is allowed:  
# quarto publish gh-pages docs
```

HTML and PDF are both declared in `_quarto.yml`. CI publishes HTML to the `gh-pages` branch; download the PDF from the book toolbar when the PDF format builds successfully.

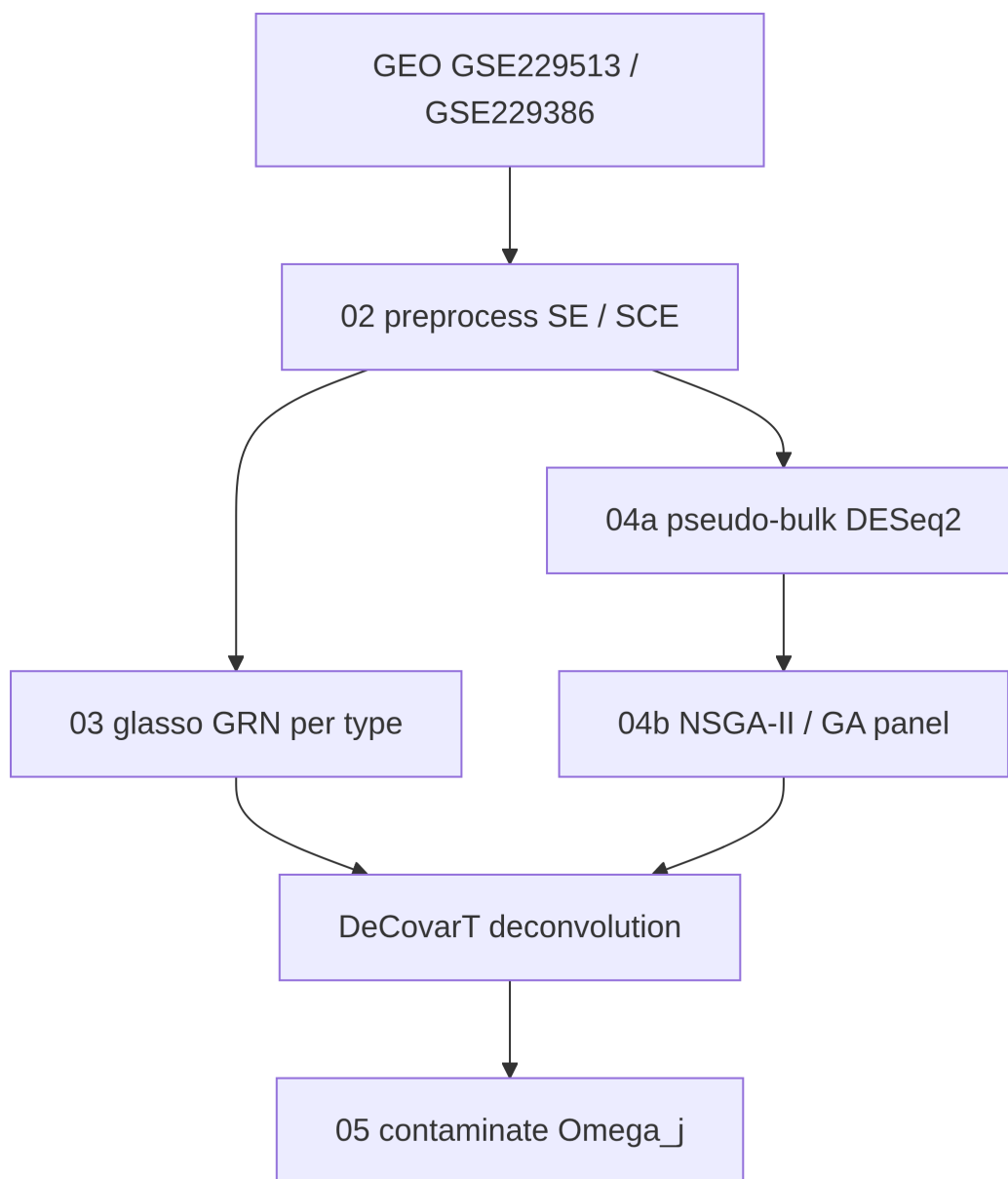


Figure 1.1: Planned pipeline: GEO gastruloids, GRN per cell type, feature panels, DeCovarT, then precision-matrix contamination.

1.2 Scripts (source of truth for computation)

Script	Role
scripts/01_01_single_cell_exploration_suppinger.R	Explore GSE229513
scripts/02_01_deconvolution_preprocessing.R	Harmonise genes; SE / SCE
scripts/03_01_grn_graphical_lasso.R	Graphical lasso per cell type
scripts/04_01_feature_selection_pseudobulk_descend.R	Select bulk markers
scripts/04_02_feature_selection_genetic_algorithm.R	Find a gene panel
scripts/05_01_sensitivity_precision_contamination.R	Specify Ω_j

2 Suppinger gastruloids

3 Data access

Single-cell object and bulk HTSeq counts from the murine gastruloid resource of Suppinger and colleagues (*Cell Stem Cell*, 2023; GEO [GSE229513](#) and [GSE229386](#)). Cite the paper from Zotero once the key exists in `bibliography/REFERENCES.bib` — do not invent a citekey.

Download commands belong in `data/raw/README.md`. Never overwrite `data/raw/`.

4 Exploration template

Port the GastroDeconv2FateMap scripts:

- [01_01_single_cell_exploration_supping.R](#)
- [02_01_deconvolution_preprocessing.R](#)

Local templates: `scripts/01_01_*.R` and `scripts/02_01_*.R`.

```
source(here::here("R", "paths.R"))  
# seurat_obj <- readRDS(path_raw("GSE229513_gastruloidsobject.rds"))
```

5 Outputs

- Intermediate `SummarizedExperiment` and `SingleCellExperiment` under `data/intermediate/`
- Codebook: `data/dictionaries/suppinger_features.csv`

6 Gene regulatory networks

7 Graphical lasso per cell type

Estimate a sparse precision matrix Ω_j for each cell type j with graphical lasso (**huge**), after gene-panel restriction.

Template: `scripts/03_01_grn_graphical_lasso.R`.

```
# huge_fit <- huge::huge(x, method = "glasso")  
# huge_sel <- huge::huge.select(huge_fit)
```

Store Ω_j and $\Sigma_j = \Omega_j^{-1}$ under `data/processed/grn/` together with λ and sparsity. These matrices are the inputs to the contamination sweep (Chapter [12](#) in the next chapters).

8 Feature selection

9 Pseudo-bulk markers

Per time point, aggregate unnormalised counts by sample and cell type, filter with `edgeR::filterByExpr()`, and fit DESeq2 as in [06_02_deseq2*.R](#).

Template: `scripts/04_01_feature_selection_pseudobulk_deseq2.R`.

10 Genetic / Pareto panels

Refine the shortlist with the all-versus-all criteria in DeCovarT `vignettes/supp-S6-feature-selection.qmd`: minimise $\kappa_2(\mu_{\mathcal{G}})$ and mean overlap with `mco::nsga2()` (NSGA-II); optional scalar GA / GenSA.

Template: `scripts/04_02_feature_selection_genetic_algorithm.R`.

```
# mco::nsga2(fn = bi_objective_func, idim = n_candidates, odim = 2, ...)
```

Keep several non-dominated panels; do not report a single arbitrary knee as *the* signature until nested, leakage-free validation exists.

11 Sensitivity analyses

12 Contaminating Ω_j

Starting from the fitted sparse precisions (Chapter 7), apply the misspecification factors described in DeCovarT `vignettes/supp-S5-misspecification.qmd`:

1. Observation noise (Gaussian, t_ν , PLN, contaminated Gaussian)
2. Missing reference component (0%, 2%, 10%)
3. Copula / PLN dependence across genes
4. Spillover between types
5. Outlier genes
6. Wrong covariance reference (perturb or subsample the glasso fit)

Template: `scripts/05_01_sensitivity_precision_contamination.R`.

Compare `DeCovarT::deconvolute_ratios()` to mean-only baselines on the **same** genes and mixtures. Pre-declare estimands and Monte Carlo error as in the package `synthetic-scenarios` vignette `{#sec-nmeth}`.